An inverse problem in oceanography: Identification of the bottom deformations from free surface measurements

PhD defense

Raphaël Terrine (UMA/ENSTA-CMAP/Ecole Polytechnique)

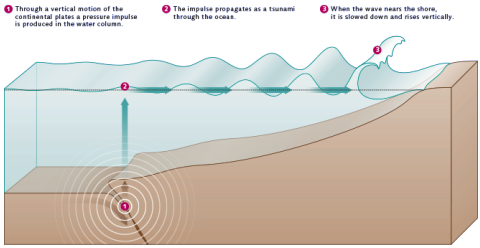
Supervisors: Laurent Bourgeois (ENSTA) and Philippe Moireau (Ecole Polytechnique)



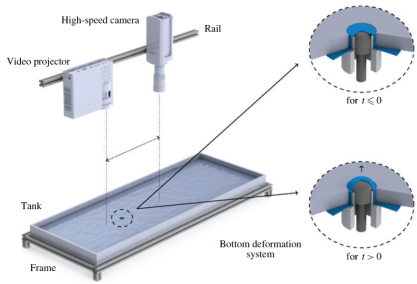




Context



(a) Formation of a tsunami

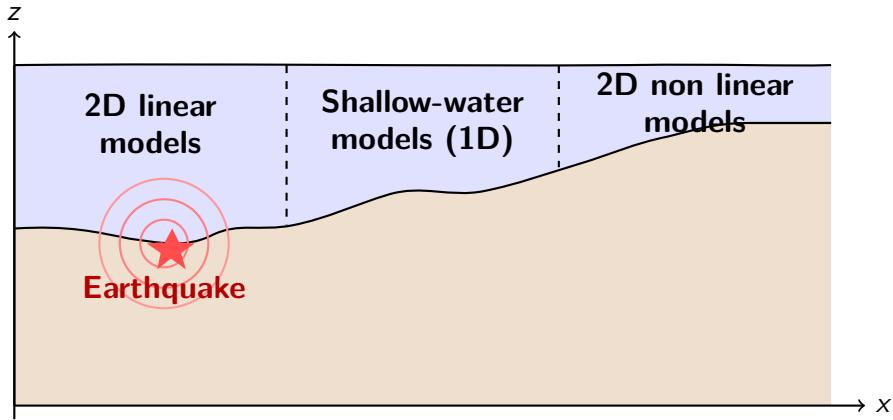


(b) Reproduce from *P.J.Cobelli et al, Determination of the bottom deformation from wave measurements, 2017.*

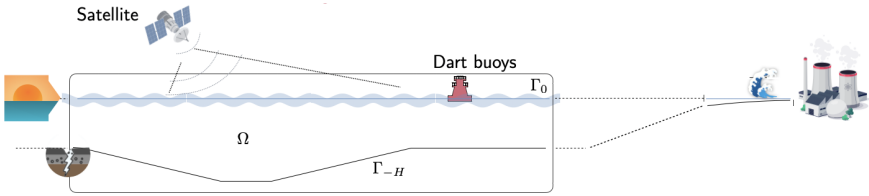
- Tsunami: seismic wave generated by the displacement of the sea bottom.
- Challenge: quickly detect the earthquake before disaster using the free surface measurements.
- Problem studied experimentally in a tank (Figure (b)).

Choice of the model in 2D: the tsunami over time

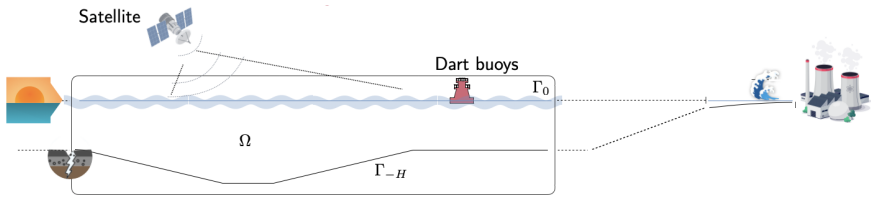
D.Dutykh, *Mathematical modelling of tsunami waves, 2007*



Complete 2D model and asymptotic limit



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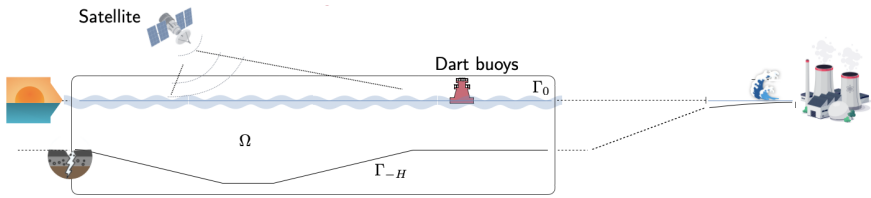
→ Obtained through linearisation of Euler equations:

$$\begin{cases} \left(\Delta - c^{-2}\partial_t^2\right)\varphi = 0 & \text{in } \Omega \times (0,T), \\ \partial_z\varphi = \partial_t s & \text{on } \Gamma_0 \times (0,T), \\ \partial_t\varphi + g s = 0 & \text{on } \Gamma_0 \times (0,T), \\ -\partial_z\varphi = -\partial_t\zeta := \chi & \text{on } \Gamma_{-H} \times (0,T), \\ \varphi(0) = \partial_t\varphi(0) = 0 & \text{in } \Omega. \end{cases}$$

📖 *J.Dubois, S.Imperiale, A.Mangeney, J.Sainte-Marie, Well-posedness and potential-based formulation for the propagation of hydro-acoustic waves and tsunamis, 2024*

- φ: velocity potential.
- s: perturbation of ocean free surface.
- χ: sea-bottom velocity.

Complete 2D model and asymptotic limit



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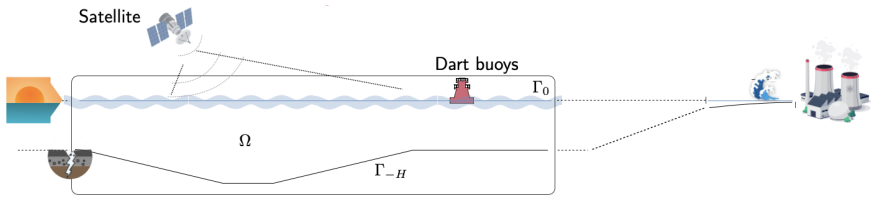
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Direct problem :

From χ, compute s .

- **Complete model**: $c < \infty$.
- **Gravity model**: $c = \infty$.

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Direct problem :

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Inverse problem

From s, retrieve χ .

Several type of solutions

$$\widetilde{W}_0(0,T) = \left\{ \varphi \in L^2(0,T;H^1(\Omega)),\ \partial_t\varphi \in L^2(0,T;L^2(\Omega)), \partial_t(\varphi|_{\Gamma_0}) \in L^2(0,T;L^2(\Gamma_0)),\ \text{and } \varphi(0)=0 \right\}$$

Variational solutions (*Section 3.3.2*)

Let $\chi \in H^1(0,T;H^{-1/2}(\Gamma_{-H}))$ with $\chi(0)=0$, then there exits $\varphi \in \widetilde{W}_0(0,T)$ a variational solution, s.t

$$\partial_t^2\varphi \in L^2(0,T;H^1(\Omega)') \text{ and } \partial_t^2(\varphi|_{\Gamma_0}) \in L^2(0,T;H^{-1/2}(\Gamma_0)),$$

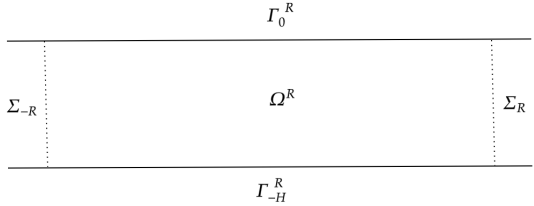
$$\text{while } \partial_t\varphi(0)=0 \text{ and } \partial_t(\varphi|_{\Gamma_0})(0)=0,$$

and for all $\tilde{\varphi} \in L^2(0,T;H^1(\Omega))$,

$$\int_0^T \frac{1}{c^2}\langle \partial_t^2\varphi, \tilde{\varphi}\rangle_{H^1(\Omega)',H^1(\Omega)} + \frac{1}{g}\left\langle \partial_t^2(\varphi|_{\Gamma_0}), \tilde{\varphi}|_{\Gamma_0}\right\rangle_{H^{-1/2}(\Gamma_0),H^{1/2}(\Gamma_0)}$$

$$+ (\nabla\varphi, \nabla\tilde{\varphi})_{L^2(\Omega)}\, dt = \int_0^T \langle \chi, \tilde{\varphi}\rangle_{H^{-1/2}(\Gamma_{-H}),H^{1/2}(\Gamma_{-H})}\, dt.$$

Reduction to a bounded domain



$$
\begin{cases}
\left(\Delta - c^{-2}\partial_t^2\right)\varphi = 0 & \text{in } \Omega^R \times (0,T), \\
\partial_y \varphi = \partial_t s & \text{on } \Gamma_0^R \times (0,T), \\
\partial_t \varphi + g s = 0 & \text{on } \Gamma_0^R \times (0,T), \\
\partial_\nu \varphi = \chi & \text{on } \Gamma_{-H}^R \times (0,T), \\
\pm \partial_x \varphi + \dfrac{1}{c}\partial_t \varphi = 0 & \text{on } \Sigma_{\pm R} \times (0,T), \\
\varphi(0) = \partial_t \varphi(0) = 0 & \text{in } \Omega^R.
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K.Dgaygui, P.Joly, Absorbing boundary conditions for the linear gravity waves, 2006

A remark on uniqueness

- Is a measurement of the free surface s associated to a unique source term χ?

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Uniqueness domain: Consequence of Holmgren Theorem

Assume that (φ, s) satisfies the model for χ. If $s = 0$ on $\Gamma_0 \times (0, T)$, then $\chi = 0$ on $\Gamma_{-H} \times (0, T - H/c)$.

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- In fact, in the asymptotic regime $c \to +\infty$, we can prove using Carleman estimates (*Section 1.5*) that we have a logarithmic reconstruction stability estimate from boundary observations to the boundary source.
  - This suggests that the problem is severely ill-posed $\to$ Low stability with respect to the noise amplitude .

L.Baudouin, M.De Buhan, S.Ervedoza,Global Carleman estimates for waves and applications, 2013

Thesis summary

1. Introduction (Chapter 1) ▶
2. Mixed formulation in the frequency domain (Chapter 2) ▶
3. One-shot mixed space-time formulation (Chapter 3) ▶
4. Boundary optimal control and iterative approach (Chapter 4) ▶
5. A link between the two approaches (Chapter 5) ▶
6. Perspectives : a Friedrich's framework (Chapter 5) ▶

1. **One-shot mixed space-time formulation (Chapter 3)**

2. **Boundary optimal control and iterative approach (Chapter 4)**

3. **Perspectives: a Friedrichs' framework (Chapter 5)**

Inverse problem formulation

Inverse problem: *For a given* $s \in L^2(0, T; L^2(\Gamma_0^R))$*, find* $\chi \in L^2(0, T; L^2(\Gamma_{-H}^R))$ *such that* $\frac{\partial \varphi}{\partial \nu} = \chi$ *with* $\varphi \in V_T$ *satisfying:*

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- Both Neumann and Dirichlet on Γ_0^R.
- Lack of conditions on Γ_{-H}^R

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Weak formulation

$\varphi \in V_T$ satisfies $\partial_t \varphi|_{\Gamma_0^R} = f$ and φ satisfies: $\forall \mu \in H^1(0,T;H^1(\Omega^R))$ such that $\mu(T) = 0$ and $\mu = 0$ on $\Gamma_{-H}^R \times (0,T)$:

$$b(\varphi,\mu) := -\frac{1}{c^2}\int_0^T \int_{\Omega^R} \partial_t \varphi \partial_t \mu \, dXdt + \int_0^T \int_{\Omega^R} \nabla\varphi.\nabla\mu \, dXdt + \frac{1}{c_R}\sum_{\pm}\int_0^T \int_{\Sigma_{\pm R}} \partial_t \varphi \mu \, dydt - \frac{1}{g}\int_0^T \int_{\Gamma_0^R} \partial_t \varphi \partial_t \mu = 0.$$

Towards an abstract framework

- $M_T = \{\mu \in H^1(0,T;H^1(\Omega^R)),\ \mu(T) = 0 \text{ in } \Omega^R\ \mu = 0 \text{ on } \Gamma^R_{-H} \times (0,T)\}$
- $B_T :\ V_T \to M_T$ such that $b(\varphi,\mu) = (B_T\varphi,\mu)_{M_T}\ \forall \mu \in M_T$.
- $C_T : V_T \to O_T := L^2(0,T;L^2(\Gamma^R_0))$ such that $C_T\varphi = \partial_t\varphi|_{\Gamma^R_0}$.
- The concatenation operator $\Lambda : \varphi \in H^1(\Omega^R) \to (B_T\varphi, C_T\varphi) \in M_T \times O_T$.

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Canonic form

With data $F = (0, f = -gs) \in M_T \times O_T$:

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Injectivity of Λ

Λ is neither injective nor onto in the complete case. Λ is injective in the asymptotic case.

The Tikhonov formulation in an abstract framework

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Tikhonov regularisation (Λ non invertible): *For $\epsilon > 0$, find $\varphi_\epsilon \in V_T$ such that:*

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Tikhonov mixed formulation

Let $\epsilon > 0$ and φ_ϵ the Tikhonov solution:
Find $(\varphi_\epsilon, \lambda_\epsilon) \in V_T \times M_T$ *such that,*

$$\begin{cases} b(v, \lambda_\epsilon) + (C\varphi_\epsilon, Cv)_{O_T} + \epsilon(\varphi_\epsilon, v)_{V_T} = (f, Cv)_{O_T} & \forall v \in V_T, \\ b(\varphi_\epsilon, \mu) - (\lambda_\epsilon, \mu)_{M_T} = 0 & \forall \mu \in M_T. \end{cases}$$

📖 E.Burman, P.Hansbo, M.G.Larson, *Solving ill-posed control problems by stabilized finite elements: an alternative to Tikhonov regularisation*, 2018

Example in the complete case

Mixed formulation in the complete case

$$
\begin{cases}
-\dfrac{1}{c^2}\displaystyle\int_0^T\int_{\Omega^R}\partial_t v\partial_t\lambda_\epsilon\ dXdt+\int_0^T\int_{\Omega^R}\nabla v.\nabla\lambda_\epsilon\ dXdt+\dfrac{1}{c}\sum_{\pm}\int_0^T\int_{\Sigma_{\pm R}}\partial_t v\lambda_\epsilon\ dydt \\
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-\dfrac{1}{c^2}\displaystyle\int_0^T\int_{\Omega^R}\partial_t\varphi_\epsilon\partial_t\mu\ dXdt+\int_0^T\int_{\Omega^R}\nabla\varphi_\epsilon.\nabla\mu\ dXdt+\dfrac{1}{c}\sum_{\pm}\int_0^T\int_{\Sigma_{\pm R}}\partial_t\varphi_\epsilon\mu\ dydt \\
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Easily discretised with usual prismatic space-time Lagrange finite elements.

The Morozov principle in the asymptotic case

- Asymptotic case : Λ injective with dense range.

Let us introduce the noise δ such that:

$$||F^{\delta} - F||_{H_T} \leq \delta < ||F^{\delta}||_{H_T}$$

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Without noise

$$\Lambda^* \Lambda \varphi_\epsilon + \epsilon \varphi_\epsilon = \Lambda^* F.$$

$$\varphi_\epsilon \xrightarrow{\epsilon \to 0} \varphi \text{ in } V_T.$$

$\Longrightarrow \epsilon \to 0$ is the optimal choice.

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Noisy data

$$\Lambda^*\Lambda\varphi_\epsilon^\delta + \epsilon\varphi_\epsilon^\delta = \Lambda^* F^\delta. \quad (1)$$

$$\varphi_\epsilon^\delta \not\xrightarrow{\epsilon \to 0} \varphi \text{ in } V_T, \ \|\varphi_\epsilon^\delta\|_{V_T} \to \infty \text{ if } F^\delta \notin Im(\Lambda).$$

$\Longrightarrow$ How to choose ϵ?

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The Morozov Principle

For φ_ϵ^δ the solution of (1), we have that there exists a unique $\epsilon(\delta)$ such that:

$$||\Lambda\varphi_\epsilon^\delta - F^\delta||_{H_T} = \delta.$$

We denote by $\varphi_\delta := \varphi_{\epsilon(\delta)}^\delta$.

How to compute φ_δ and $\epsilon(\delta)$?

- **The dual problem:** I.Ekeland, R.Temam, *Convex analysis and variational problems, 1999*

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× L^* Non quadratic.

The quadratisation of the problem

P.Demeestère, A remark on the relation between the Tykhonov regularization and constraint relaxation for an optimal control problem, 1998

$$L^*(q) = \frac{1}{2}\|\Lambda^* q\|^2_{V_T} + \delta\|q\|_{H_T} - (q, F^\delta)_{H_T} \rightarrow L^*_\epsilon(q) = \frac{1}{2}\|\Lambda^* q\|^2_{V_T} + \frac{1}{2}\epsilon\|q\|^2_{H_T} - (q, F^\delta)_{H_T}.$$

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Computation of $\epsilon(\delta)$ through the following algorithm:

- Let $\epsilon_k > 0$.
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$$q_k \xrightarrow{k\to\infty} q_\delta \text{ in } H_T, \quad \Lambda^* q_k \xrightarrow{k\to\infty} \varphi_\delta \text{ in } V_T, \quad \epsilon_k \xrightarrow{k\to\infty} \epsilon(\delta).$$

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✓Linear variational formulation at each iteration.

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✓Linear variational formulation at each iteration.
✓Fast convergence with respect to k (10 iterations).

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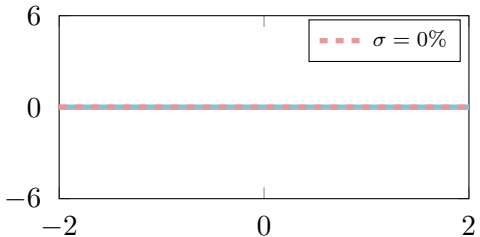
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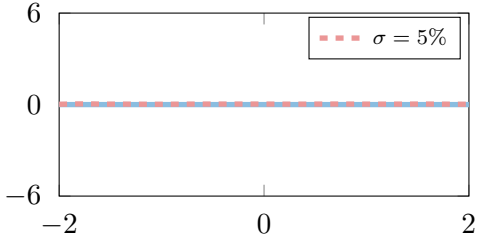
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✓Linear variational formulation at each iteration. × Huge space-time matrices.
✓Fast convergence with respect to k (10 iterations).

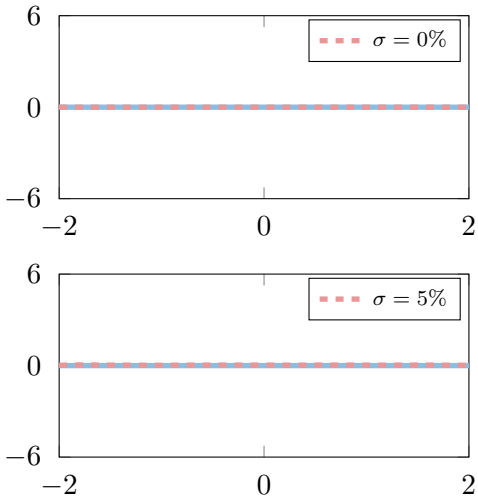
Numerical results: complete model



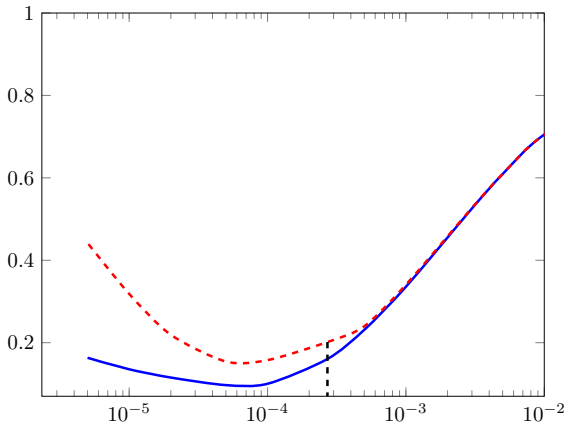


$h = 0.15$ and $\tau = 0.01$.

Numerical results: complete model



$h = 0.15$ and $\tau = 0.01$.



$\epsilon \mapsto \|\chi_\epsilon^\delta - \chi\|$ for $\sigma = 0\%$ and $\sigma = 5\%$.

Conclusion on the space-time mixed formulation

Pros:

- One-shot approach.
- Rigorous method to compute the regularisation parameter $\epsilon(\delta)$.
- Strong stability with respect to the noise.

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Cons :

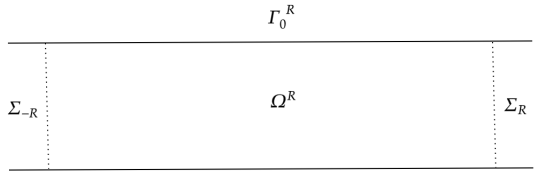
- Too large matrix to invert $\rightarrow$ Space-time (Use of iterative SOR method)
- Difficult extension to 3D in space.

1. One-shot mixed space-time formulation (Chapter 3)

2. **Boundary optimal control and iterative approach (Chapter 4)**

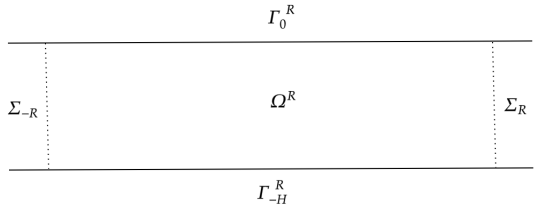
3. Perspectives: a Friedrichs' framework (Chapter 5)

Parabolic regularization



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\begin{cases}
\left(\Delta - c^{-2}\partial_t^2\right)\varphi = 0 & \text{in } \Omega^R \times (0,T), \\
\partial_\nu \varphi = \partial_t s & \text{on } \Gamma_0^R \times (0,T), \\
\partial_t \varphi + g s = 0 & \text{on } \Gamma_0^R \times (0,T), \\
\partial_\nu \varphi = \chi & \text{on } \Gamma_{-H}^R \times (0,T), \\
\pm\, \partial_x \varphi + \dfrac{1}{c}\partial_t \varphi = 0 & \text{on } \Sigma_{\pm R} \times (0,T), \\
\varphi(0) = \partial_t \varphi(0) = 0 & \text{in } \Omega^R.
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J-L.Lions, E.Magenes, Non homogeneous boundary value problems and applications, 1972
J-L.Lions,Optimal Control of Systems Governed by Partial Differential Equations,1971

Let V, H, U three Hilbert spaces such that $V \hookrightarrow H \equiv H' \hookrightarrow V'$.

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$$W_0(0,T) = \left\{ z \in L^2(0,T;V),\ \dot{z} \in L^2(0,T;V'),\ z(0) = 0 \right\}.$$

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- Source term $\chi \in L^2(0,T;U)$.
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Theorem

The problem has a unique solution z and:

$$\|z\|_{W_0(0,T)} \leq c \|\chi\|_{L^2(0,T;U)}.$$

The complete cas as an order 1 system

- $H = H^1(\Omega^R) \times L^2(\Omega^R) \times L^2(\Gamma_0^R)$ with the following norm

$$\|(\varphi, \psi, s)^T\|_H^2 = \|\varphi\|_{H^1(\Omega^R)}^2 + \frac{1}{c^2}\|\psi\|_{L^2(\Omega^R)}^2 + g\|s\|_{L^2(\Gamma_0^R)}^2,$$

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The variational first order system

The regularised problem solution $z = (\varphi, \psi, s)^T \in W_0(0, T)$ is a solution in the variational sense of:

$$\dot{z} + A_\eta z = B\chi \text{ in } L^2(0, T; V').$$

Optimal control approach for solving the inverse problem

Objective

Recovering χ through the measurement y of s in $L^2(0, T, L^2(\Gamma_0^R))$.

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Tikhonov regularisation: For $\epsilon > 0$:

$$\inf_{\chi \in L^2(0,T;L^2(\Gamma_{-H}^R))} \mathcal{J}(\chi) = \int_0^T \frac{1}{2}\|y(t) - Cz(t)\|^2_{L^2(\Gamma_0^R)} + \frac{\epsilon}{2}\|\chi(t)\|^2_{L^2(\Gamma_{-H}^R)}\, dt,$$

with $\forall z = (\varphi, \psi, s)^T \in W_0(0, T)$, $Cz = s$ and

$$\begin{cases} \dot{z} + A_\eta = B\chi, \\ z(0) = 0, \end{cases}$$

J-L.Lions,*Optimal Control of Systems Governed by Partial Differential Equations,1971*

Optimal control approach for solving the inverse problem : abstract framework

Let Y, V, H, U four Hilbert spaces with $V \subset H \subset V'$. Let $C \in \mathcal{L}(H, Y)$: measurement operator.

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Recover χ through the measurement y of Cz.

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Well-posedness

The previous problem has a unique solution : $\mathcal{J}$ continuous, coercive, strictly convex.

Discretization of the method

$$\mathcal{J}_\tau^h\left(\{\chi_h^n\}_{1\leq n\leq N}\right) = \sum_{n=0}^{N} \frac{\tau}{2}\|y_h^n - C_h z_h^n\|_Y^2 + \sum_{n=1}^{N} \frac{\tau\epsilon}{2}\|\chi_h^n\|_U^2,$$

$$\text{with } \begin{cases} \dfrac{z_h^n - z_h^{n-1}}{\tau} + A_h z_h^n = B_h \chi_h^n, \quad \forall n \in [\![1; N]\!] \\ z_h^0 = 0. \end{cases}$$

F-X.Le Dimet,Optimal control for data assimilation in meteorology,1990

Discretization of the method

$$\mathcal{J}_\tau^h\left(\{\chi_h^n\}_{1\leq n\leq N}\right) = \sum_{n=0}^{N} \frac{\tau}{2}\|y_h^n - C_h z_h^n\|_Y^2 + \sum_{n=1}^{N} \frac{\tau\epsilon}{2}\|\chi_h^n\|_U^2,$$

$$\text{with } \begin{cases} \dfrac{z_h^n - z_h^{n-1}}{\tau} + A_h z_h^n = B_h \chi_h^n, \quad \forall n \in [\![1; N]\!] \\ z_h^0 = 0. \end{cases}$$

Optimality conditions:

$$\begin{cases} \dfrac{z_h^n - z_h^{n-1}}{\tau} + A_h z_h^n = B_h \chi_h^n, \quad \forall n \in [\![1; N]\!], \\ \dfrac{p_h^n - p_h^{n+1}}{\tau} + A_h^* p_h^n = C_h^* \left(y_h^n - C_h z_h^n\right), \ \forall n \in [\![1; N]\!], \\ \epsilon\chi_h^n - B_h^* p_h^n = 0, \ \forall n \in [\![1; N]\!], \\ z_h^0 = p_h^{N+1} = 0. \end{cases}$$

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Optimality conditions:

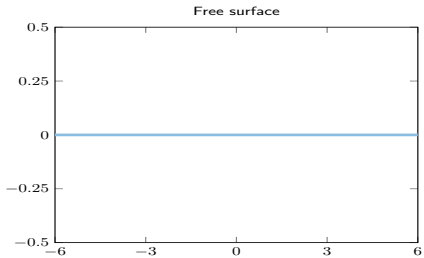
$$\begin{cases} \dfrac{z_h^n - z_h^{n-1}}{\tau} + A_h z_h^n = B_h \chi_h^n, \quad \forall n \in [\![1; N]\!], \\ \dfrac{p_h^n - p_h^{n+1}}{\tau} + A_h^* p_h^n = C_h^*\left(y_h^n - C_h z_h^n\right), \ \forall n \in [\![1; N]\!], \\ \epsilon\chi_h^n - B_h^* p_h^n = 0, \ \forall n \in [\![1; N]\!], \\ z_h^0 = p_h^{N+1} = 0. \end{cases}$$

Solving the problem as an engineer

- Compute analytically the gradient of $\mathcal{J}_\tau^h$.
- Use a gradient descent algorithm to find the minimiser.
- We use L-BFGS for fast convergence.

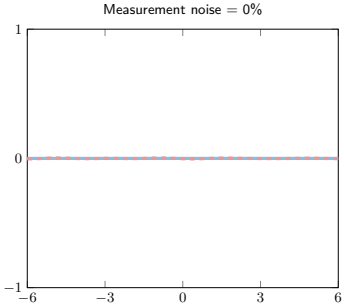
F-X.Le Dimet,Optimal control for data assimilation in meteorology,1990

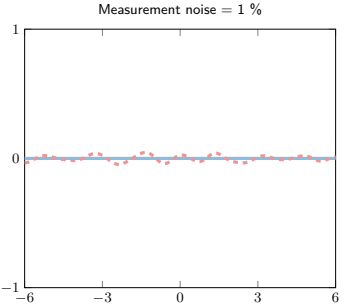
Numerical reconstruction

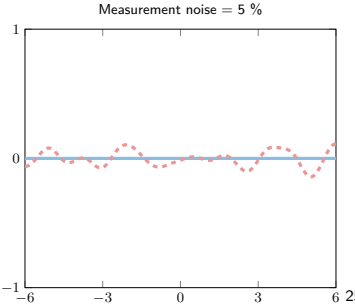


Parameters

- $R = 90cm.$
- $H = 15cm.$
- $h = 0.1, \mathbb{P}_2$ elements.
- $\tau = h^2.$
- $\epsilon = \nu = 10^{-6}.$







Our contribution: Numerical analysis of the scheme

$$\inf_{\chi \in L^2(0,T;U)} \mathcal{J}(\chi) = \int_0^T \frac{1}{2}\|y(t) - Cz(t)\|_Y^2 + \frac{\epsilon}{2}\|\chi(t)\|_U^2 \, dt,$$

with z the solution in $W_0(0,T)$ of $\dot{z} + A_\eta z = B\chi$.

Our contribution: Numerical analysis of the scheme

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Questions

- $\{\chi_h^n\}_{1\le n\le N} \to \overline{\chi}$ when $(\tau, h) \to 0$?
- Rate of convergence ?

An auxiliary mixed formulation

Find $Z = (z, \chi)^T \in W_0(0,T) \times L^2(0,T,U)$ and $p \in L^2(0,T;V)$ such that for all $\tilde{Z} = (\tilde{z}, \tilde{\chi})^T \in W_0(0,T) \times L^2(0,T,U)$ and for all $\tilde{p} \in L^2(0,T;V)$:

$$\begin{cases} j(Z, \tilde{Z}) + d(\tilde{Z}, p) = \ell(\tilde{Z}), \\ d(Z, \tilde{p}) = 0, \end{cases}$$

$$\left| \begin{array}{l} j(Z, \tilde{Z}) = \displaystyle\int_0^T \epsilon(\chi, \tilde{\chi})_U + (Cz, C\tilde{z})_Y \; dt, \\ d(Z, p) = \displaystyle\int_0^T \langle \dot{z}, p \rangle_{V',V} + a_\eta(z, p) - b(\chi, p) \; dt, \\ \ell(Z) = \displaystyle\int_0^T (y, Cz)_Y \; dt. \end{array} \right.$$

Well-posedness of the mixed formulation (*Section 4.3*)

Coercivity of j on $\ker d$

There exists $C > 0$ such that for all $Z \in W_0(0, T) \times L^2(0, T; U)$ such that $d(Z, p) = 0, \forall p \in L^2(0, T; V)$, we get:

$$j(Z, Z) \geq C \|Z\|^2_{W_0(0,T) \times L^2(0,T;U)}$$

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$$j(Z,Z) \geq C\|Z\|^2_{W_0(0,T)\times L^2(0,T;U)}$$

Inf-sup condition

There exists $\beta > 0$ such that:

$$\inf_{p\in L^2(0,T;V)} \sup_{Z\in W_0(0,T)\times L^2(0,T;U)} \frac{d(Z,p)}{\|Z\|_{W_0(0,T)\times L^2(0,T;U)}\|p\|_{L^2(0,T;V)}} \geq \beta.$$

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Fundamental result: Equivalence between problems

The solution χ of the optimal control problem coincides with the solution χ of the auxiliary mixed formulation.

A.Ern,J-L Guermond,Theory and Practice of Finite Elements,2004

Finite elements discretisation

- **Discretisation spaces:** $V_h \subset V$, $U_h \subset U$ et $Y_h \subset Y$ for $h > 0$.
- Let $W_h^\tau = \{w_h^\tau \in \mathbb{P}_1 \otimes V_h, w_h^\tau(0) = 0\}$, $V_h^\tau = \mathbb{P}_0(0,T) \otimes V_h$, $U_h^\tau = \mathbb{P}_0(0,T) \otimes U_h$

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Discrete mixed formulation

Find $Z_h^\tau = (z_h^\tau, \chi_h^\tau)^T \in W_h^\tau \times U_h^\tau$ and $p_h^\tau \in V_h^\tau$ such that for all $\tilde{Z}_h^\tau = (\tilde{z}_h^\tau, \tilde{\chi}_h^\tau)^T \in W_h^\tau \times U_h^\tau$ and for all $\tilde{p}_h^\tau \in V_h^\tau$:

$$\begin{cases} j(Z_h^\tau, \tilde{Z}_h^\tau) + d(\tilde{Z}_h^\tau, p_h^\tau) = \ell(\tilde{Z}_h^\tau), \\ d(Z_h^\tau, \tilde{p}_h^\tau) = 0, \end{cases}$$

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Discrete mixed formulation

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$$z_h^\tau(x,t) = \sum_{n=1}^{N} z_h^n(x)\omega^n(t);\ \chi_h^\tau(x,t) = \sum_{n=1}^{N} \chi_h^n(x)\omega_0^n(t);\ y_h^\tau(x,t) = \sum_{n=1}^{N} y_h^n(x)\omega_0^n(t), p_h^\tau(x,t) = \sum_{n=1}^{N} p_h^n(x)\omega_0^n(t).$$

Conforming time scheme

Time scheme with conforming FEM

$$
\begin{cases}
\dfrac{z_h^n - z_h^{n-1}}{\tau} + A_h \dfrac{z_h^n + z_h^{n-1}}{2} = B_h \chi_h^n, \quad \forall n \in [\![1; N]\!], \\
\dfrac{p_h^n - p_h^{n+1}}{\tau} + A_h^*\left(\dfrac{p_h^{n+1} + p_h^n}{2}\right) = C_h^* \left(\dfrac{y_h^{n-1} + y_h^n}{2} - C_h \dfrac{z_h^{n-1} + 4z_h^n + z_h^{n+1}}{6} \right), \ \forall n \in [\![1; N-1]\!] \\
\dfrac{p_h^N - p_h^{N+1}}{\tau} + A_h^*\left(\dfrac{p_h^{N+1} + p_h^N}{2}\right) = C_h^* \left(\dfrac{y_h^{N-1} + y_h^N}{2} - C_h\left(\dfrac{1}{6} z_h^{N-1} + \dfrac{1}{3} z_h^N\right) \right), \\
\epsilon \chi_h^n - B_h^* p_h^n = 0, \quad \forall n \in [\![1; N]\!], \\
p_h^{N+1} = z_h^0 = 0.
\end{cases}
$$

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Time scheme with conforming FEM

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\dfrac{p_h^n - p_h^{n+1}}{\tau} + A_h^*\left(\dfrac{p_h^{n+1} + p_h^n}{2}\right) = C_h^* \left(\dfrac{y_h^{n-1} + y_h^n}{2} - C_h \dfrac{z_h^{n-1} + 4z_h^n + z_h^{n+1}}{6} \right), \ \forall n \in [\![1; N-1]\!] \\
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\epsilon \chi_h^n - B_h^* p_h^n = 0, \quad \forall n \in [\![1; N]\!], \\
p_h^{N+1} = z_h^0 = 0.
\end{cases}
$$

- Crank-Nicholson scheme rather than implicit Euler scheme (no uniform energy identity).
- Data attachment term different from that of the Euler implicit scheme.

Adding a quadrature rule

With $\fint_0^T f(t)\ dt = \tau \sum_{n=1}^{N} f(n\tau_-)$:

- $j(Z_h^\tau, \tilde{Z}_h^\tau) \to j_\tau(Z_h^\tau, \tilde{Z}_h^\tau) = \int_0^T \epsilon(\partial_t \chi_h^\tau, \partial_t \tilde{\chi}_h^\tau)_U\ dt + \fint_0^T (C_h z_h^\tau, C_h \tilde{z}_h^\tau)_Y\ dt.$
- $\ell(\tilde{Z}_h^\tau) \to \ell_\tau(\tilde{Z}_h^\tau) = \fint_0^T (y_h^\tau, C_h \tilde{z}_h^\tau)_Y\ dt.$
- $d(Z_h^\tau, \tilde{p}_h^\tau) \to d_\tau(Z_h^\tau, \tilde{p}_h^\tau) = \int_0^T \langle \dot{z}_h^\tau, \tilde{p}_h^\tau \rangle_{V',V} - b(\chi_h^\tau, \tilde{p}_h^\tau)\ dt + \fint_0^T a_h(z_h^\tau, \tilde{p}_h^\tau)\ dt.$

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Time scheme with quadrature rule

$$
\begin{cases}
\dfrac{z_h^n - z_h^{n-1}}{\tau} + A_h z_h^n = B_h \chi_h^n, \quad \forall n \in [\![1; N]\!], \\
\dfrac{p_h^n - p_h^{n+1}}{\tau} + A_h^* p_h^n = C_h^* \left(y_h^n - C_h z_h^n\right), \ \forall n \in [\![1; N]\!] \\
\epsilon \chi_h^n - B_h^* p_h^n = 0 \quad \forall n \in [\![1; N]\!], \\
p_h^{N+1} = z_h^0 = 0.
\end{cases}
$$

Well-posedness of the discrete problem

Discrete mixed formulation with quadrature

Find $Z_h^\tau = (z_h^\tau, \chi_h^\tau)^T \in W_h^\tau \times U_h^\tau$ and $p_h^\tau \in V_h^\tau$ such that for all $\tilde{Z}_h^\tau = (\tilde{z}_h^\tau, \tilde{\chi}_h^\tau)^T \in W_h^\tau \times U_h^\tau$ and for all $\tilde{p}_h^\tau \in V_h^\tau$:

$$\begin{cases} j_\tau(Z_h^\tau, \tilde{Z}_h^\tau) + d_\tau(\tilde{Z}_h^\tau, p_h^\tau) = \ell_\tau(\tilde{Z}_h^\tau), \\ d_\tau(Z_h^\tau, \tilde{p}_h^\tau) = 0. \end{cases}$$

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- j_τ coercive on $\ker d_\tau$ with a constant that depends on $\|P_h\|_{\mathcal{L}(V)}$.
- Inf-sup condition for d_τ with constant that depends on $\|P_h\|_{\mathcal{L}(V)}$.

Technical tool

Let us introduce the H-orthogonal projection onto V_h: 📖 *A.Ern,J-L Guermond,Finite Elements III,2021*

$$P_h : V \to V_h, \;\; P_h(z) = \underset{z_h \in V_h}{\arg\min} \|z - z_h\|_H.$$

Well-posedness of the discrete problem

Discrete mixed formulation with quadrature

Find $Z_h^\tau = (z_h^\tau, \chi_h^\tau)^T \in W_h^\tau \times U_h^\tau$ and $p_h^\tau \in V_h^\tau$ such that for all $\tilde{Z}_h^\tau = (\tilde{z}_h^\tau, \tilde{\chi}_h^\tau)^T \in W_h^\tau \times U_h^\tau$ and for all $\tilde{p}_h^\tau \in V_h^\tau$:

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The discrete mixed formulation is well-posed. 📖 *F. Brezzi, M. Fortin, Mixed and hybrid element methods, 1991*

Well-posedness of the discrete problem

Discrete mixed formulation with quadrature

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The discrete mixed formulation is well-posed. 📖 *F.Brezzi,M.Fortin,Mixed and hybrid element methods,1991*

- **Idea of the proof** (*Section 4.6*): Energy identity of the discrete problem with a control on the discrete norms : $\|z_h^\tau\|_V^2 + \|\dot{z}_h^\tau\|_{V_h'}^2 \leq C\|\chi_h^\tau\|_U^2$.
- We introduced the norm $\|v_h\|_{V_h'} := \sup_{w_h \in V_h} \frac{\left| <v_h, w_h>_{V',V} \right|}{\|w_h\|_V}$. 📖 *A.Ern,J-L Guermond,Finite Elements III,2021*
- We have then for all $v_h \in V_h$, $\|v_h\|_{V_h'} \leq \|v_h\|_{V'} \leq \|P_h\|_{\mathcal{L}(\mathcal{V})}\|v_h\|_{V_h'} \rightarrow$ control on the continuous norms.

Convergence rate

Second Strang Lemma (Proposition 4.6.5)

$$\|Z_h^\tau - Z\|_{W_0(0,T)\times L^2(0,T;U)} \le C\|P_h\|^3_{\mathcal{L}(V)} \Bigg(\inf_{Z_h^\tau \in W_h^\tau \times U_h^\tau} \|Z - Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}$$

$$+ \inf_{Z_h^\tau \in W_h^\tau \times U_h^\tau} \|Z - Z_h^\tau\|_{L^\infty(0,T;V)\times L^2(0,T;U)} + \inf_{p_h^\tau \in V_h^\tau} \|p - p_h^\tau\|_{L^\infty(0,T;V)}$$

$$+ \sup_{\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \frac{|j(Z,\tilde{Z}_h^\tau) - j_\tau(Z,\tilde{Z}_h^\tau)|}{\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}} + \sup_{\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \frac{|d(\tilde{Z}_h^\tau, p) - d_\tau(\tilde{Z}_h^\tau, p)|}{\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}}$$

$$+ \sup_{\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \frac{|\ell(\tilde{Z}_h^\tau) - \ell_\tau(\tilde{Z}_h^\tau)|}{\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times H_l^1(0,T;U)}} + \sup_{\tilde{p}_h^\tau \in V_h^\tau} \frac{|d(Z,\tilde{p}_h^\tau) - d_\tau(Z,\tilde{p}_h^\tau)|}{\|\tilde{p}_h^\tau\|_{L^2(0,T;V)}} \Bigg)$$

Convergence rate

Second Strang Lemma (Proposition 4.6.5)

$$\|Z_h^\tau - Z\|_{W_0(0,T)\times L^2(0,T;U)} \leq C\|P_h\|^3_{\mathcal{L}(V)} \Bigg(\inf_{Z_h^\tau \in W_h^\tau \times U_h^\tau} \|Z - Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}$$

$$+ \inf_{Z_h^\tau \in W_h^\tau \times U_h^\tau} \|Z - Z_h^\tau\|_{L^\infty(0,T;V)\times L^2(0,T;U)} + \inf_{p_h^\tau \in V_h^\tau} \|p - p_h^\tau\|_{L^\infty(0,T;V)}$$

$$+ \sup_{\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \frac{|j(Z,\tilde{Z}_h^\tau) - j_\tau(Z,\tilde{Z}_h^\tau)|}{\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}} + \sup_{\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \frac{|d(\tilde{Z}_h^\tau,p) - d_\tau(\tilde{Z}_h^\tau,p)|}{\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}}$$

$$+ \sup_{\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \frac{|\ell(\tilde{Z}_h^\tau) - \ell_\tau(\tilde{Z}_h^\tau)|}{\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times H_l^1(0,T;U)}} + \sup_{\tilde{p}_h^\tau \in V_h^\tau} \frac{|d(Z,\tilde{p}_h^\tau) - d_\tau(Z,\tilde{p}_h^\tau)|}{\|\tilde{p}_h^\tau\|_{L^2(0,T;V)}} \Bigg)$$

Convergence error for $\mathbb{P}_k$ Lagrange finite elements

$$\sqrt{\|z_h^\tau - z\|^2_{W_0(0,T)} + \|\chi_h^\tau - \chi\|^2_{L^2(0,T;U)}} \leq C\|P_h\|^3_{\mathcal{L}(V)}(\tau + h^k)$$

Consequence for the complete tsunami model

Theorem (Proposition 4.7.1)

$$\|P_h\|_{\mathcal{L}(V)} \leq 1/\sqrt{h}$$

Mesh assumption : quasi-uniform mesh

We suppose that the mesh is "shape regular" and there exists $c > 0$ such that:

$$\forall h > 0,\ \forall K \in \mathcal{T}_h,\ ch \leq h_K := Diam(K) \leq h.$$

Consequence for the complete tsunami model

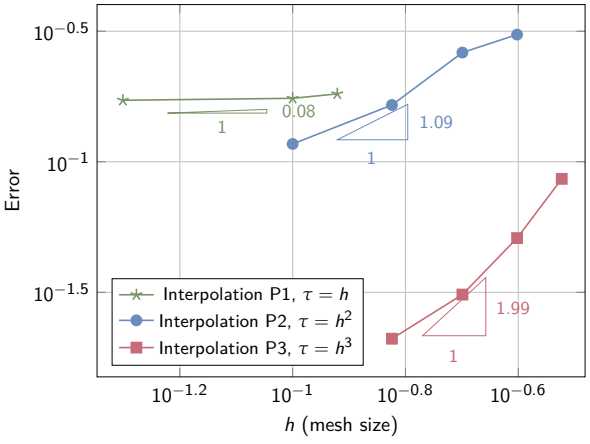
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- Parameters $H = 0.15, R = 0.9, T = 1.$
- Expected errors

$$\|\chi_h^\tau - \chi\|_{L^2((0,T)\times\Gamma^R_{-H})} \leq \frac{C_{st}}{h^{\frac{3}{2}}}(\tau + h^k)$$

- when choosing $\tau = h^k$
  - if $k = 1$, no convergence
  - if $k = 2$, order $1/2$
  - if $k = 3$, order $3/2$



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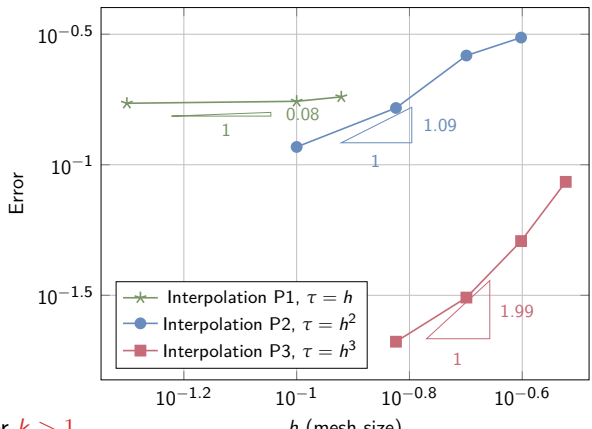
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Numerically, we have better convergence rates for $k > 1$

1. One-shot mixed space-time formulation (Chapter 3)

2. Boundary optimal control and iterative approach (Chapter 4)

3. **Perspectives: a Friedrichs' framework (Chapter 5)**

Perspectives without dissipation

- Joint work with J. Dubois and S. Impériale

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- We rewrite the initial model as a Friedrichs' system:

$$\begin{cases} \partial_t U - G^* V = 0 \\ \partial_t V + GU = B \displaystyle\int_0^t \chi(s) ds \text{ where } B \ : \ H^{-\frac{1}{2}}(\Gamma_{-H+\theta}) \to \mathcal{G} \text{ s.t. } (B\chi, (u,v))_{\mathcal{G}} = \langle \chi, u \rangle \\ U(0) = V(0) = 0 \end{cases}$$

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- Discrete inf-sup condition $\rightarrow$ Work to be done .

Efficient discretization with high-order spatial finite elements and an explicit scheme will not satisfy a discrete inf-sup condition

High order explicit discretization approach

$$\mathcal{J}_\tau^h\left(\{\chi_h^n\}_{1\le n\le N}\right) = \sum_{n=0}^{N}\frac{\tau}{2}\|y_h^n - \gamma_{\Gamma_0}\varphi_h^n\|^2_{L^2(\Gamma_0)} + \sum_{n=1}^{N}\frac{\tau\epsilon}{2}\|\chi_h^n\|^2_{L^2(\Gamma_{-H+\theta})},$$

$$\text{with } \begin{cases} \dfrac{U_h^{n+1} - 2U_h^n + U_h^{n-1}}{\tau^2} + A_h U_h^n + D_h\dfrac{U_h^{n+1} - U_h^{n-1}}{2\tau} = B_h\chi_h^n, \quad \forall n \in [\![0; N-1]\!] \\ U_h^0 = U_h^{-1} = 0, \end{cases}$$

with $U_h^n = (\varphi_h^n, \gamma_{\Gamma_0}\varphi_h^n)^T$, $A_h = G_h G_h^*$. and D_h absorbing boudary conditions operator.

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- Adding Gauss-Lobatto quadrature rule → Explicit scheme : No inversion required

G.Haine,K.Ramdani,Reconstructing initial data using observers: error analysis of the semi-discrete and fully discrete approximations,2012

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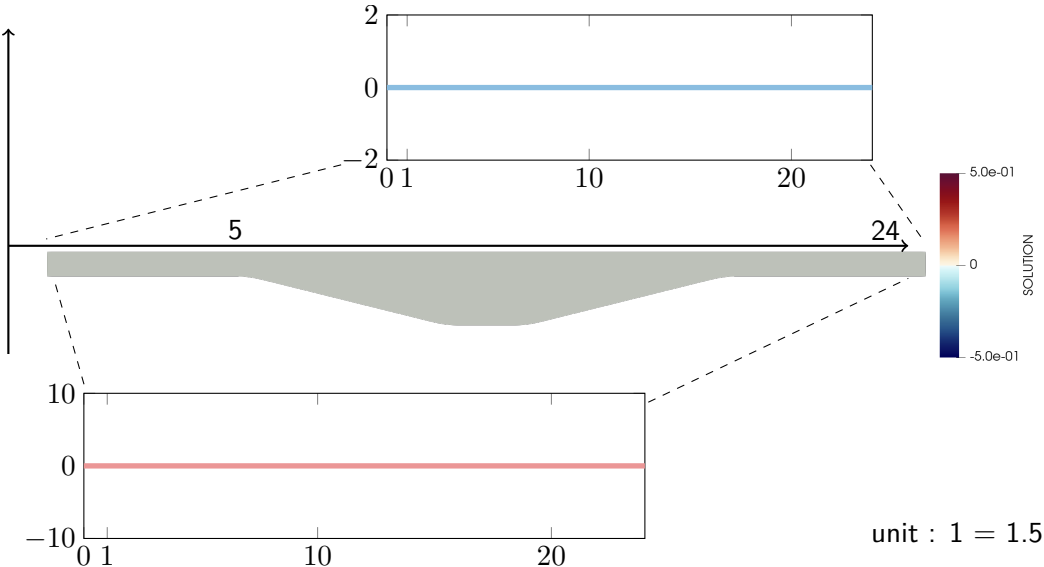
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- Adding Gauss-Lobatto quadrature rule → Explicit scheme : No inversion required
- High order space discretization: $\mathbb{Q}_4$ finite elements.

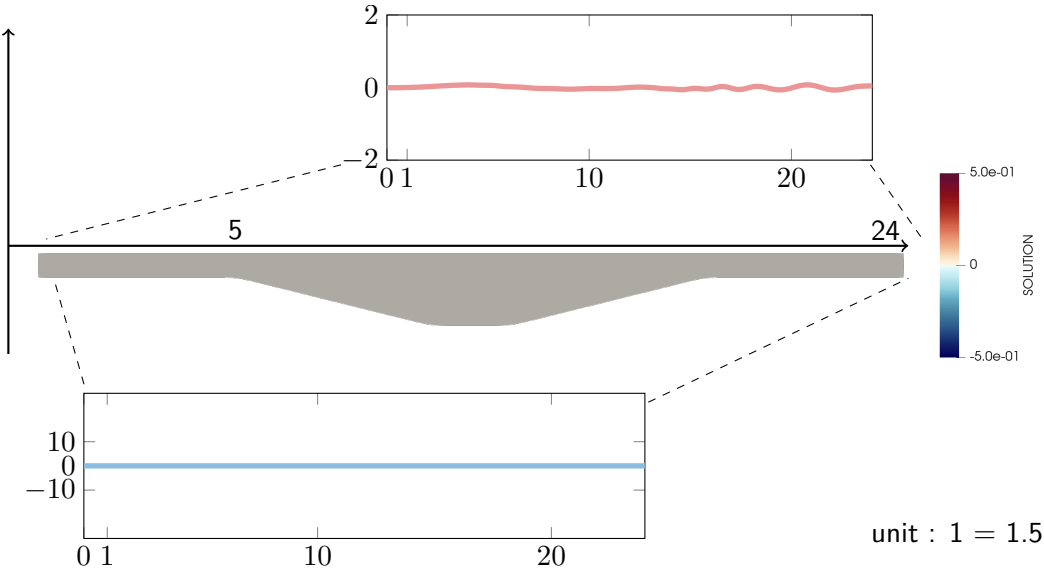
G.Haine,K.Ramdani,Reconstructing initial data using observers: error analysis of the semi-discrete and fully discrete approximations,2012

Forward problem



unit : 1 = 1.5 km

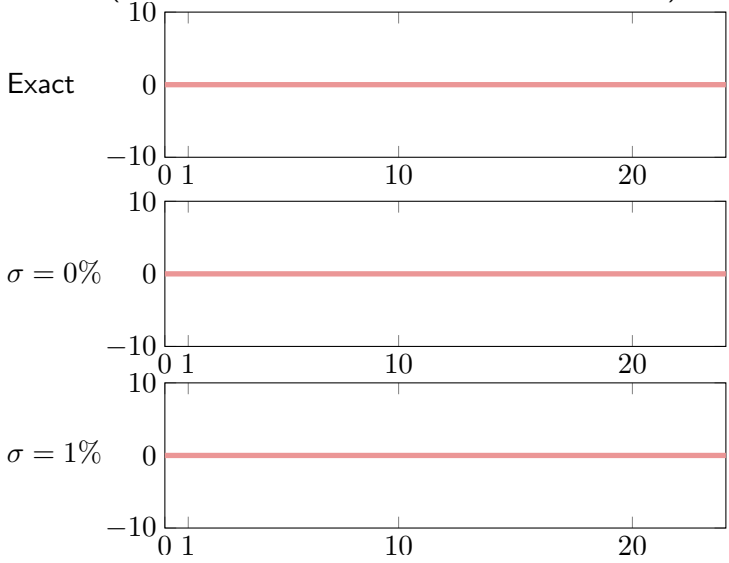
Adjoint problem



unit : 1 = 1.5 km

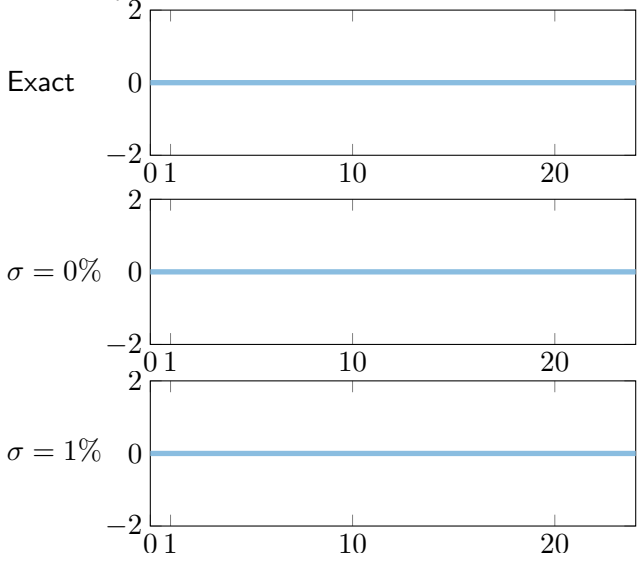
Reconstruction of the bottom source

- Reconstruction: (321 dof for the source, 4000 BFGS iterations) – Slow-down visualization



Associated free surface: prevision in the future

- Reconstruction: (321 dof for the source, 4000 BFGS iterations)



Conclusion

Conclusion:

- Tsunami identification in two mathematical distincts formalism : monolithical space-time mixed formulation and iterative optimal control method.
- Distincts but similar → See Chapter 5 of the thesis.

Perspectives:

- Solve the inverse problem thanks to a Kalman filter→ Monitoring the free surface.
- Discrete inf-sup condition in the Friedrichs' framework → Numerical analysis.
- Bathymetry problem in the time domain.
- Duality approach to choose the regularisation parameter in the iterative approach.
- Real data reconstructions.

Thank you for your attention

Choice of ϵ with respect to the noise level

